ThaliSense

An on-device AI nutrition & metabolic-health coach built for Indian meals

2nd LaserHack 2026 · Round 1: Idea Submission
Theme: **HealthTech** (with Artificial Intelligence)

THEME	STAGE	RUNS ON	COST TO USER
HealthTech + AI	Working MVP, live	Any phone browser	Free, no sign-up

1 · PROBLEM STATEMENT

101 M
Indians living with diabetes (ICMR-INDIAB, Lancet D&E 2023)

136 M
more with prediabetes, most of them undiagnosed

BMI 23
is where overweight starts for Asian Indians, not 25

India's diabetes and obesity epidemic is driven by everyday food: large portions of white rice, refined flour, fried snacks and sweet chai. Yet the tools people have do not fit them:

- **Food apps are built for Western diets.** They don't know *pesarattu*, *bisi bele bath* or *kadhi*, and can't understand “do roti aur dahi” or “rendu dosai”.
- **Risk is mis-measured.** South Asians develop diabetes at a lower body weight and with more belly fat, so global BMI cut-offs reassure people who are already at risk.
- **Generic advice fails.** “Eat less rice” doesn't stick. Swaps for foods people already eat, in their own regional cuisine and within their budget, do.
- **Privacy and cost.** Many users won't upload meal photos and health data to a server, or pay for a dietitian.

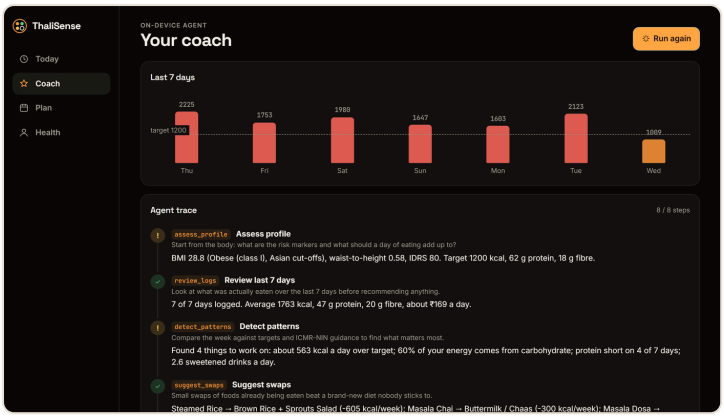
2 · PROPOSED SOLUTION

ThaliSense is a free web app that turns what people already eat into a personal, explainable plan. Everything, including the AI, runs inside the user's browser.

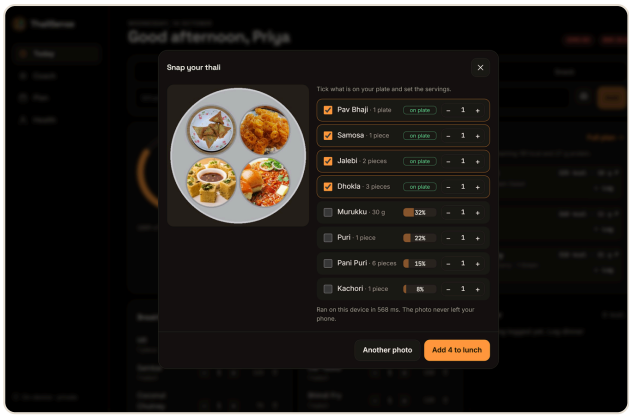
- **Log by typing, in your own words.** A meal parser covering 108 Indian dishes understands quantities, Hindi and Tamil number words, regional names, plurals and typos.
- **Log by photo, even a whole thali.** An on-device vision model (87.8% top-1, 95% top-3 on 1,694 test photos) scans the plate for every dish and suggests the usual sides. One tap confirms.
- **Know your real risk.** The validated **Indian Diabetes Risk Score** (MDRF), BMI with **Asian Indian cut-offs**, and waist-to-height ratio, with clear advice to get tested when the risk is high.
- **An explainable coach agent.** It reviews the past week, finds the biggest problems (carb-heavy plates, low protein, sugary drinks), and proposes swaps with the weekly calorie, protein and fibre impact of each.
- **A self-checking 7-day plan.** Regional menus (South, North, East, West; veg, egg or non-veg) within a ₹/day budget. The agent verifies every day against the targets and repairs any misses, and every step is shown to the user.
- **Private, installable, offline.** No account and no server: data stays on the phone, and it installs like an app.

UNIQUE FEATURES

WHAT	WHY IT IS DIFFERENT
Indian-first food intelligence	108 dishes with IFCT-2017-based nutrition, GI band, regional tags and rupee cost; 60 real meal templates instead of random food combinations.
On-device vision AI, fine-tuned for Indian food	4 models benchmarked; MobileCLIP-S2 chosen and few-shot adapted on 2,354 labelled photos: 71.5% → 87.8% top-1 vs the CLIP baseline, in a smaller 72 MB download. Runs on WebGPU or multi-threaded WASM, offline after first use.
Transparent agent, not a black box	Eight traceable tool steps (assess → review → detect → swap → plan → verify → budget → habits). It never invents numbers.
Clinically grounded	MDRF IDRS, Asian BMI consensus (Misra 2009), Mifflin-St Jeor, ICMR-NIN 2024 dietary guidelines.



The coach agent's visible reasoning trace and 7-day intake chart.



On-device dish recognition with suggested side dishes.

3 · TECHNOLOGY STACK

`Typed meal` → `parser` · `Photo` → `MobileCLIP-S2 (WebGPU / WASM worker)` → `meal log (on device)` → `coach agent` → `insights · swaps · plan`

LAYER	CHOICE
Front end	React 19, TypeScript, Vite; responsive, works on low-end Android browsers; dark and light themes
AI / ML	Transformers.js on ONNX Runtime Web; MobileCLIP-S2 (fp16, 72 MB, cached); 108 few-shot-adapted dish vectors (76 KB); 5-region thali scan
Agent & logic	Deterministic tool-based planner in TypeScript: template scoring, portion scaling, verify-and-repair loop, budget check
Data	IFCT 2017 (ICMR-NIN) based food table; IDRS, Asian BMI and ICMR-NIN 2024 rules
Quality & hosting	17 unit + 12 end-to-end tests incl. a WCAG 2 AA accessibility scan, GitHub Actions CI, Vercel, installable PWA. No backend and no paid APIs.

4 · REAL-WORLD IMPACT & FEASIBILITY

Impact

- **Earlier detection.** IDRS flags high-risk people (score of 60 or more) and pushes them to a cheap fasting-glucose or HbA1c test.
- **Measurable change.** On the demo week the coach finds swaps worth about 1,200 kcal a week (around 0.7 kg of fat a month) while raising protein and fibre.
- **Inclusive.** Regional, vegetarian-aware and budget-aware (default ₹200 a day), in the words people actually use.
- **Scales to the public system.** An ASHA-worker mode could screen whole villages offline on one phone.

Feasibility

- **Already working.** Live with typed and photo logging, thali detection, risk scoring, progress tracking, the coach agent and 7-day plans.
- **Near-zero running cost.** Static hosting; all compute happens on users' devices, so a million users cost the same as ten.
- **Measured accuracy.** 87.8% top-1 and 95.0% top-3 on 1,694 photos from three datasets; whole-thali scanning doubles the dishes found.
- **Next (Round 2).** Plate segmentation for crowded thalis, Hindi and Tamil voice logging, and an ASHA-worker screening mode.

ThaliSense is a wellness guide, not a medical device. It does not diagnose; it encourages users at high risk to get tested.

Live MVP: multihacks-unstop26.vercel.app Code: github.com/andringodson/Hackathon-MultiHacksUnstop26